COUNTABLE PRODUCTS AND KERNELS OF SPLIT MAPS MAKE A PREADDITIVE CATEGORY KAROUBIAN

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the first assertion; the closing sentence 'The proof of (2) is dual (using the dual definition of a Karoubian category, namely condition (2) of Lemma lemma-karoubian).' is the author's own duality remark on the part written out in full inside the counted span, and it is preserved verbatim.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Karoubian categories. Same chapter, Lemma lemma-karoubian. Supplies the two equivalent descriptions of a Karoubian category, in particular the dual description invoked for the second assertion of the proof.
- Kernels as representing objects. Same chapter, the definition of a kernel and its description by a representable functor. Supplies the observation that the functor sending an object W to the kernel of the induced endomorphism of the Hom-set is representable exactly when the idempotent has a kernel.

Source reference. Original lemma 09SH: sources/stacks-homology/pinned-original-source.tex, source lines 743–743. Navigation pointer invoked at proof line 803; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{D}$ be a preadditive category.*

(1) *If $\mathcal{D}$ has countable products and kernels of maps which have a right inverse, then $\mathcal{D}$ is Karoubian.*

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(2) *If $\mathcal{D}$ has countable coproducts and cokernels of maps which have a left inverse, then $\mathcal{D}$ is Karoubian.*

Proof. Let X be an object of $\mathcal{D}$ and let $e : X \to X$ be an idempotent. The functor

$$W \longmapsto \mathrm{Ker}(\mathrm{Mor}_{\mathcal{D}}(W, X) \xrightarrow{e} \mathrm{Mor}_{\mathcal{D}}(W, X))$$

is representable if and only if e has a kernel. Note that for any abelian group A and idempotent endomorphism $e : A \to A$ we have

$$\mathrm{Ker}(e : A \to A) = \mathrm{Ker}(\Phi : \prod\nolimits_{n \in \mathbf{N}} A \to \prod\nolimits_{n \in \mathbf{N}} A)$$

where

$$\Phi(a_1, a_2, a_3, \ldots) = (ea_1 + (1 - e)a_2, ea_2 + (1 - e)a_3, \ldots)$$

Moreover, Φ has the right inverse

$$\Psi(a_1, a_2, a_3, \ldots) = (a_1, (1 - e)a_1 + ea_2, (1 - e)a_2 + ea_3, \ldots).$$

Hence (1) holds. The proof of (2) is dual (using the dual definition of a Karoubian category, namely condition (2) of Lemma 09SH). □